

Parkinson's Disease

Introduction

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- **Parkinson's Disease (PD)** is a **slowly progressive, neurodegenerative disease** of extrapyramidal motor system that features classic motor symptoms of **tremor at rest, rigidity, bradykinesia/akinesia, and postural/gait instability** (TRAP)
- Symptoms seen when **~70–80%** of dopaminergic neurons in the substantia nigra are lost, a profound loss of dopamine (DA) in the striatum

Epidemiology

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- The Average age of onset is **around 60**
- 10 per 100,000 persons between 50–59 years of age) to 120 per 100,000 persons between 80–89 years of age)
- Prevalence of PD increase with age
 - Less frequent in patient less than 50
 - 1% in patients >65 yrs
 - 2.5% in those >80yrs
- A higher incidence is reported among men
 - **male: female ratio of 2:1**
- Ethiopia has the world's lowest recorded prevalence of PD

Etiology

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- **Unknown etiology**

Possible suggested causes

1. Environmental factors(extrinsic)

- Neurotoxins that are highly selective to substantia nigra dopaminergic neurons: **1-methyl-4-phenyl-1,2,3,6-tetrahydropyridine (MPTP)**
 - Several pesticides have molecules structurally similar to MPTP

Chronic exposure to pesticides and heavy metals (such as iron and manganese), rural living, and drinking well water

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2. Dopamine metabolism can generate free radicals through auto oxidation & MAO metabolism

❖ Controversial but possible if the normal antioxidative mechanism is overwhelmed

3. Genetic factors

- If the disease occur before age 50
- mutation of *α -synuclein* & *parkin* genes

4. Aging process

Pathophysiology

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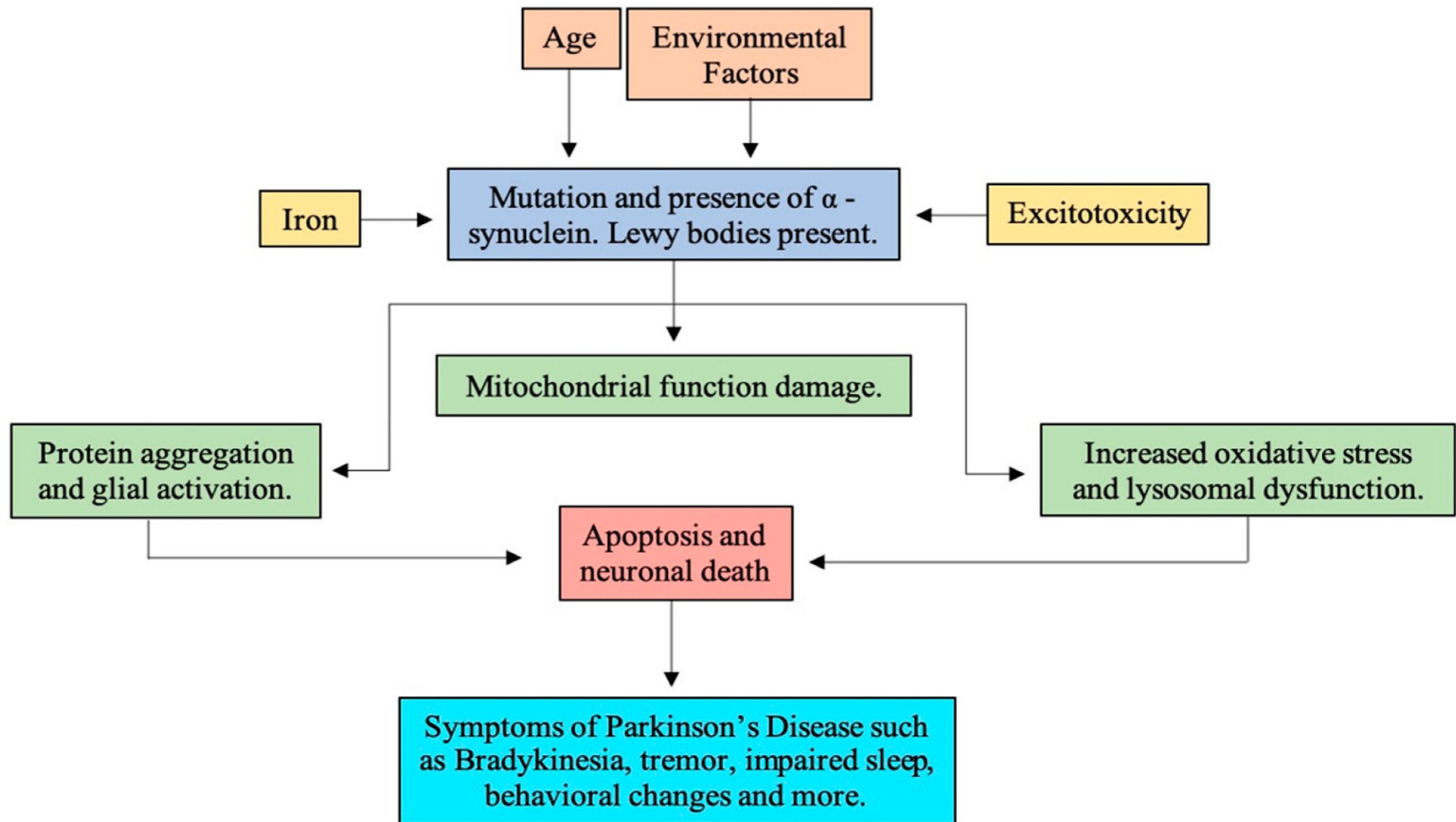
Two **histological hallmarks** in PD

- **Loss of dopamine producing neurons in Substantia Nigra (SN)**
- **Presence of lewy bodies in the remaining neurons**
 - Lewy bodies consist of largely **mutated** *α -synuclein* (synaptic protein of unknown function)
 - Mutation render this protein resistance to degradation & accumulate: abnormal aggregates of protein that develop inside nerve cells
 - Possibly increase susceptibility to oxidative attack

Con't

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- In PD, the striatum portion of the basal ganglia receives an inadequate amount of nigra cells, which impairs a person's ability to control movement.
- Imbalance of dopamine and acetylcholine in the striatum occurs
 - Clinically detectable PD occurs when 70 to 80% of dopaminergic production in the substantia nigra are lost



Clinical Presentation

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- *Patients with PD display both **motor** and **non-motor** symptoms*
- **Motor Symptoms (mnemonic TRAP)**
 - T**=Tremor at rest (“pill rolling”)
 - R** = Rigidity (stiffness and rigidity)
 - A** = Akinesia or bradykinesia
 - P** = Postural instability and gait abnormalities
- Diagnostic criteria: **at least two** of the above should be present
- The motor manifestations usually began unilaterally but later became bilateral as the disease progress

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Tremor at rest: Involves the upper extremities and often is the sole compliant

- however **only 2/3 of PD** patients have tremor at diagnosis
- Usually occurs in the **hands or arms** with characteristic 'pill-rolling' motion
 - Can also occur in head, face, jaw, & leg
- Usually unilateral but may progress to become bilateral & worsens with stress
- **Disappears with purposeful movement such as picking up an object and absent during sleep**

Rigidity

- Increased muscular resistance to passive range of motion and commonly affect the upper & lower extremities
- If tremor occurs at the affected extremities, Can also affect the face resulting in Hypomimia

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Bradykinesia

- Refers to the **slowness of movement**
- Characteristic problems
 - Slowness in carrying out any action and difficulty in initiation of mov't
 - Slowness in movement performance
 - Rapid fatigue during movement
 - Impair tasks such as handwriting
 - Intermittent immobility(freezing), especially walking through a narrow door way or taking a turn

Characteristic Problems

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- **Micrographia**: small handwriting
- **Hypomimia**: decreased facial animation (eye blinking)
- **Hypophonia**: reduced voice volume
- **Dysarthria**: slurred speech

Today is a sunny day in California

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“Today is a sunny day in California”

Postural instability

- Postural instability is a result of the loss of reflexes necessary to maintain balance when ambulating and stooped posture
- **Most common in advance stage of PD and also the most disabling**
 - High risk of fall and **generally resistance to treatment**

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Non-motor symptoms (mnemonic SOAP)

S= Sleep disturbances (insomnia, rapid eye movement sleep behavioral disorder, restless legs syndrome)

O= Other miscellaneous symptoms (problems with nausea, fatigue, speech, pain, vision)

A= Autonomic symptoms (drooling, constipation, sexual dysfunction, urinary problems, sweating, orthostatic hypotension, dysphagia)

P= Psychological symptoms (anxiety, psychosis, cognitive impairment, depression)

Stages of PD 5 stages

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Stage	Symptoms
One	• Unilateral
Two	• Bilateral • No balance impairment
Three	• Balance impairment • Mild to moderate disease • Physically independent
Four	• Severe disability • Still able to walk & stand unassisted
Five	• Wheelchair-bound or bedridden unless assisted

Diagnosis

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- **At least two of the following:** limb muscle rigidity, resting tremor, Bradykinesia or postural instability.
- Need to eliminate secondary causes:
 - **Drug-Induced**
 - Antipsychotics, Antiemetic's (metoclopramide, prochlorperazine)
 - **Toxic**
 - CO poisoning, hydrogen sulfide, manganese, methanol, MPTP
 - Stroke ,Trauma or Neoplasm involving the nigrostrial pathway
 - **Other neurodegenerative conditions:** Alzheimer's.....

Management

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Goals of therapy

- Improve motor and non-motor symptoms
- Activities of daily living (ADLs)
- Increase QOL by alleviating the patient's symptoms
- Minimizing the development of response fluctuations
- Limiting medication-related adverse effects

Non-Pharmacological Therapy

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- It includes education, support, and lifestyle modifications such as exercise
- Maintaining good nutrition, physical condition, and social interaction
- Dietary modifications improve constipation, nausea, erratic drug absorption, and minimize the risk of aspiration and weight loss

Pharmacologic Therapy

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- Available pharmacologic therapies only give **symptomatic relief**
 - Currently there is no drug that prevent degeneration of dopaminergic neuron

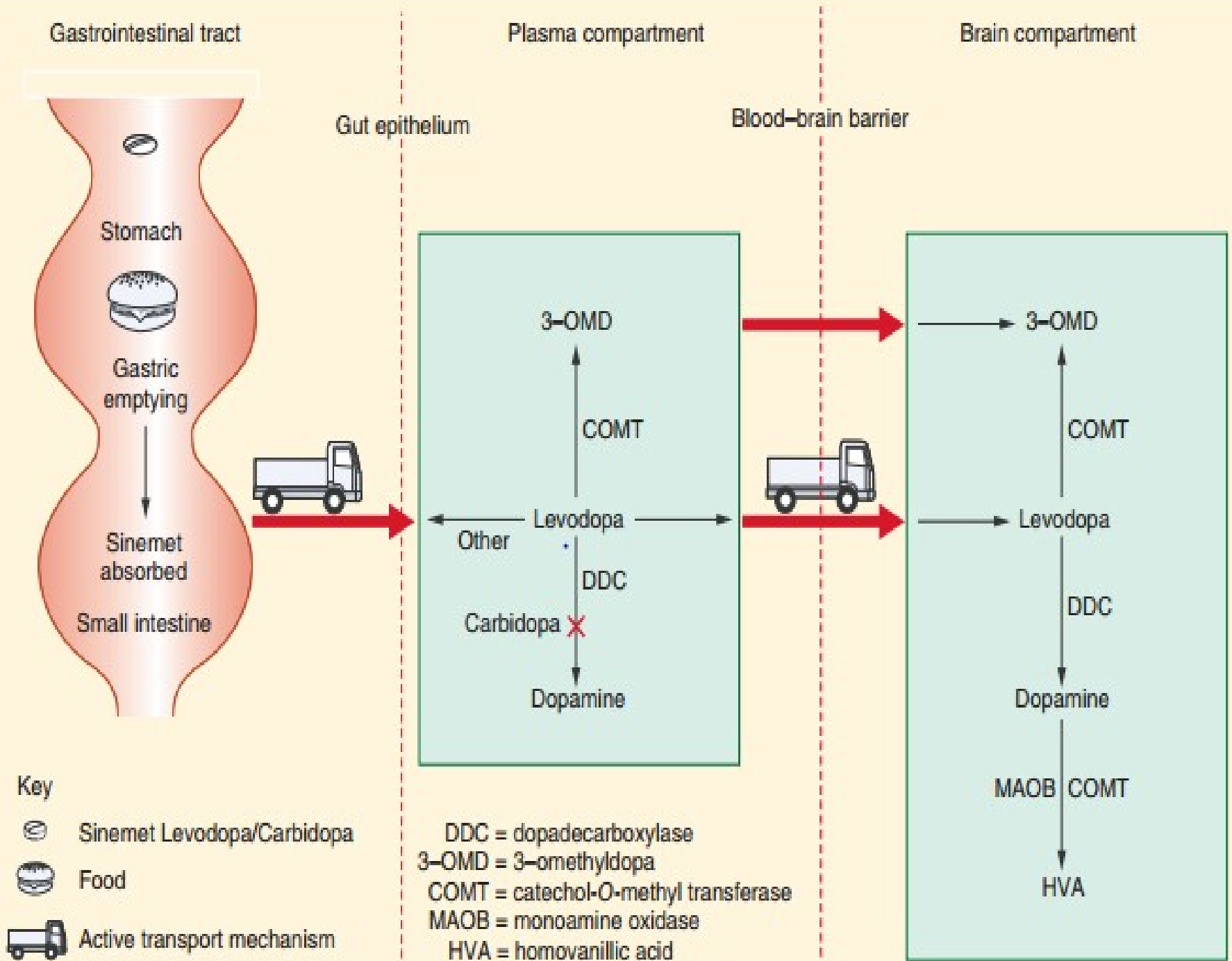
Class of Drugs Used

- **Levodopa/carbidopa (Sinemet)**
- Dopamine agonists
- Amantadine
- COMT inhibitors
- MAO-B inhibitors
- Anticholinergics

L-dopa/Carbidopa

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- L-dopa is an immediate precursor of dopamine
- In combination with carbidopa (peripherally acting L-amino acid decarboxylase inhibitor)
 - Remain the most effective drug for symptomatic relief of IPD-'gold standard'
- The role of carbidopa is prevention of peripheral conversion of L-dopa to dopamine
 - Increased amount of L-dopa will be delivered to brain plus minimize peripheral unwanted effects
- Regardless of what the initial therapeutic agent is, ultimately all patients with PD will require L-dopa at some point (WHY???)



Dose & Side Effects

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- Initial maintenance dose 25/100 mg TID(Carbidopa/L-dopa)
- About 75 mg/day is required to sufficiently inhibit the peripheral activity of L-amino acid decarboxylase
- As the disease progresses, higher doses are used with 800 to 1000mg/day as the maximum.
- An orally disintegrating tablet is available for those patients unable to swallow
- **Emergent side effects**
 - Nausea & vomiting, anorexia, postural hypotension, sedation, confusion, vivid dreams, urine discoloration.....
 - Suggestions to minimize
 - Slow build up of dose(increment 100mg/wk)

Motor Complications of L-Dopa

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- Long term administration of L-dopa is associated with a variety of **motor complication**
- Motor complications can occur as early as 6 months after starting L-dopa therapy, especially if excessive doses are used initially

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1. End-of-dose wearing off (motor fluctuation)

- Off-poor movement(return of tremor, rigidity or slowness)
- On-good movement(normal)
- Return to a poor movement state before the next dose is given
- Occur due to:
 - Increase loss of neuronal storage capacity of dopamine(progressive loss of neurons)
 - Short half-life of L-dopa
- As a result, patients become more dependent on exogenous L-DOPA
 - Pharmacokinetics of L-DOPA will be a determining factor

Possible treatments: Increase frequency of L-DOPA, add MAOI or COMT or dopamine agonist, add or switch to extended-release carbidopa/L-dopa

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2. “Delayed-on” or “No-on” response

- It's a condition where administration of L-dopa either produce a delayed effect or not at all
 - due to **delayed gastric emptying or decrease absorption in the duodenum**
- **Possible treatment:**
 - give carbidopa/L-dopa on empty stomach or
 - oral disintegrating tablet or
 - use apomorphine subcutaneously-rescue therapy

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3. Dyskinesia

- Involuntary mov't involving the neck, trunk, lower & upper extremities
- Associated with large dose of L-dopa
 - Too much dopamine activity at the striatum
- **Possible solution**
 - lower dose of L-dopa/carbidopa or
 - add amantadine

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4. Start hesitation (“freezing”)

- ▣ Increase carbidopa/L-dopa dose
- ▣ Add a dopamine agonist or MAO-B inhibitor
- ▣ Utilize physical therapy along with assistive walking devices

Anticholinergic Agents

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- DA provides negative feedback to Ach neurons in the striatum, and degeneration of **nigrostriatal dopamine neurons** also results in a relative increase of striatal cholinergic interneuron activity
 - Increased cholinergic activity is believed to contribute to the **tremor of PD**.
- Considered effective against tremor but not more so than dopaminergic agents
- Can be used as monotherapy or **adjuncts**

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- The use is limited because of intolerable side effects especially in elderly
- Possible adverse effects include:
 - dry mouth, blurred vision, somnolence, hallucinations, memory impairment, confusion, urinary retention, and constipation.
- Agents-
 - Trihexyphenidyl : 2 mg and 5 mg
 - Benztropine: 0.5 – 1 mg with a maximum of 6 mg
- Younger patients are better able to tolerate anticholinergic side effects, whereas,
 - drug class is avoided in patients with advanced age, pre-existing cognitive deficits, and dysphagia.

Amantadine

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- Possible mechanism suggested
 - Dopaminergic & non dopaminergic(inhibition of glutamate receptors)
- Modest symptomatic relief
- It is most often used for management of L-dopa-induced dyskinesia.
 - Possibly due to its anti-glutamate effect
- Dose: 300mg/d in divided dose

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- **Caution in renal failure patients**
 - **Crcl: 30-50ml/min-----100mg/d**
15-29ml/min-----100mg every other day
<15ml/min-----200 every 7 days
- **Unpleasant side effects such as**
 - **Dizziness, confusion, hallucinations, dry mouth, nightmares, nausea, peripheral edema, and livedo reticularis**

MAO-B Inhibitors

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- Two agents available
 - Selegiline and Rasagiline
 - As a class two major problem associated with MAO-B inhibitors:
 - Interaction with food(cheese rxn) and drugs(ephedrine, phenylephedrine, pseudoephedrine)
 - Transient hypertension & headache
 - Cheese contain tyramine(indirect sympathomimetics)
 1. Additive sympathomimetic effect
 2. Tyramine is a substrate of MAO found in gut and liver
- Note: large amount of tyramine(>400mg) is required for the rxn to occur because of their selectivity to MAO-B
- Serotonin syndrome
 - Concomitant use with drugs that enhance serotonin activity(mepridine, SSRI)

Selegiline

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- **Monotherapy or adjunct**
- When used as monotherapy show modest improvement in motor fluctuation
- When used as adjunctive therapy, extends the **'on' time of L-dopa up to 1hr**
- Minimal unwanted effects: Insomnia (especially if administered at bedtime), hallucinations, and jitteriness
 - Also **Potentiate L-dopa induced dyskinesia and psychiatric symptoms**
- Dosage : 5 mg BID with breakfast and lunch
 - : 1.25-2.5mg once daily as oral disintegrating tablet
 - Transmucosal absorption bypass metabolism
- Controversial theory of **decreased rate of neuronal death** due to a reduction of free radicals.

Rasagiline

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- Effective as **monotherapy in early PD and also for managing motor fluctuations in advanced PD**
 - Early initiation(perhaps even before the onset of functional impairment) is associated with better long-term outcomes
- For patients with motor fluctuation appear to **extend the 'on' time by 1hr**

When an adjunctive agent is required for managing motor fluctuations, rasagiline is considered a first-line agent apart from entacapone

Dopamine Agonists

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- Two subtypes
 - Ergot derivatives: bromocriptine & Pergolide
 - Non ergot derivatives: Pramipexol, ropinirole, rotigotine
- The non- ergot derivatives:
 - safer, more effective for mild to moderate PD and as an adjunct to L-dopa in patients with motor fluctuation
- Use of bromocriptine & pergolide has fallen now a days
 - bromocriptine- pulmonary fibrosis
 - Pergolide-valvular heart disease

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- Place of therapy: Monotherapy or combination
- Are particularly useful for:
 - Prolonging the effective treatment period of L-dopa in patients with deteriorating response
 - In younger patients with mild PD dopamine agonists are preferred due to minimal motor fluctuation associated with L-dopa
 - Treating patients who cannot tolerate high doses of L-dopa.
- Associated with more side effects than L-dopa especially in elderly: L-dopa preferred
- Potential adverse effects: somnolence, dyskinesia, nausea, vomiting, orthostatic hypotension, nightmares, hallucinations, confusion, dizziness

Ergot Agonist Dosing

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- Bromocriptine (Parlodel)

- Initial 1.25mg QD-BID
- Titrate 1.25mg to 2.5mg/d every week
- Average dose <30mg/day.
- Some patients may require up to 120mg/day

- Pergolide (Permax)

- 13 times more potent than bromocriptine
- Initial 0.05mg/d for 2 days, increasing by 0.1 to 0.15mg/d every 3 days over a 12 day period
- May increase by 0.25mg every 3 days until symptoms are eliminated or adverse effects occur
- Mean dose 3mg/d

Pramipexole

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- Initial dose - 0.125 mg TID
 - Increased every 5 to 7 days as tolerated up to 3 to 4.5mg/d
- Mean 27% reduction of L-Dopa dose
- Decrease dose inrenal function impairment
- Drugs that are secreted by the cationic transport system may decrease the clearance of pramipexole by 20%
 - Cimetidine, diltiazem, quinidine, quinine, ranitidine, triamterene, and verapamil.

Ropinirole

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- **Monotherapy or Adjunct**
- **Initial dose -- 0.25mg TID**
 - Increased by 0.25mg TID on a weekly basis to a max of 24mg/d
- **Mean 19% reduction of L-dopa dose**
- **Drugs that inhibit or induce CYP1A2 will affect the clearance of ropinirole.**
 - Inhibitors such as cimetidine, ciprofloxacin, clarithromycin, diltiazem, enoxacin, erythromycin, fluvoxamine, mexiletine, norfloxacin, omeprazole, ritonavir, and troleandomycin.
 - Inducers such as carbamazepine, phenobarbital, phenytoin, and rifampin.

COMT Inhibitors

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- Two agents:
 - Entacapone
 - Tolcapone
- MOA-prevent the peripheral conversion of L-dopa to dopamine, increasing its central bioavailability
- Place of therapy
 - As an add on to L-dopa for extending its effect & managing “wearing off”
 - Alone- no effect on PD
- COMT inhibition is more effective than controlled release L-dopa/carbidopa in the management of ‘wearing off’

Entacapone

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- Adjunct therapy
- Initial dose of 200mg with each dose of levodopa up to 8 times daily due to its **short half life**
- **Chosen over tolcapone when an add on is needed because of no hepatotoxicity**
- Decrease dose of L-dopa may be necessary
 - Exacerbation of L-dopa side effects such as diarrhea, urine discoloration, abdominal pain

Tolcapone

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- Adjunct therapy
- Initial 100mg TID up to 200mg TID
- **More potent and longer acting** than entacapone
- Exacerbation of L-dopa side effects such as diarrhea, urine discoloration
 - Decrease L-dopa dose.....by 25 to 50%
- **Fatal liver toxicity** limits its use
 - **Monitor LFTs**
 - every 2 weeks for 1 year,
 - every 4 weeks for 6 months, then
 - every 8 weeks
- **Reserved** for patients with fluctuations that are not responding to other therapies

Special situations

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- ❑ **Hallucinations or psychosis**
- ✓ **Discontinue or reduce** anti-Parkinson medications as tolerated.
- ✓ Then **start with anticholinergics**; then use MAO-B inhibitors, amantadine, dopamine agonists, COMT inhibitors; finally, use carbidopa/ levodopa
- ✓ Pimavanserin
- ✓ Quetiapine or Clozapine
- ✓ Avoid FGA, Risperidone & Olanzapine –worsen motor symptoms

Special situations

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- Cognitive disorders
- ✓ **Discontinue or reduce** anti-Parkinson medications as tolerated
- ✓ **Rivastigmine** or other cholinesterase inhibitors
- Sleep disorders, depression, agitation, anxiety, constipation, orthostatic hypotension, seborrhea, blepharitis, and restless legs syndrome can occur in Parkinson disease; treat as usual.

Cases

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1. L.T. takes carbidopa/levodopa 25 mg/100 mg orally four times daily and trihexyphenidyl 2 mg orally three times daily for Parkinson's disease. L.T.'s wife reports that his movements are very slow and that he is having trouble walking. 16. Given these symptoms,

Which change seems most reasonable?

- A. Increase carbidopa/levodopa, discontinue trihexyphenidyl.
- B. Continue carbidopa/levodopa, increase trihexyphenidyl.
- C. Decrease carbidopa/levodopa, continue trihexyphenidyl.
- D. Decrease carbidopa/levodopa, increase trihexyphenidyl.

Cases

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2. Six months later, L.T. returns to the clinic, concerned that his carbidopa/levodopa dose is wearing off before his next dose is due, because his tremor and slow movements are worse before the next dose of carbidopa/ levodopa.

Which recommendation is best?

- A. Increase the carbidopa/levodopa dose.
- B. Decrease the carbidopa/levodopa dose.
- C. Increase the dosing interval.
- D. Decrease the dosing interval.

THE END